

Code No: 217AD

KR21**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Supplementary Examinations, December-2025/January-2026

DATA VISUALIZATION TECHNIQUES

(COMMON TO CSE, CSE (AI&ML), CSE (DS))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(20 Marks)

1. a) What is Data Visualization.

[2M]L1

b) Differentiate Data Visualization and Computer Graphics.

[2M]L2

c) Write short note on Bertins Semiology of Graphics.

[2M]L1

d) Why is contrast important in visualization?

[2M]L2

e) Write short notes on Probing Two-Dimensional Data

[2M]L1

f) Explain Slicing techniques in Visualizing Volume Data.

[2M]L2

g) What are the advantages of visualizing a single document using techniques like word clouds or tag clouds?

[2M]L2

h) List any four Interaction Techniques.

[2M]L1

i) What is the importance of information density in designing a visualization?

[2M]L1

j) How do limitations in human perception influence visualization design?

[2M]L2

PART- B

(5 X 10 =50 Marks)

2. a) How data visualization is related to computer science and information technology?

[5M]L2

b) Explain the significance of handling missing data and describe common techniques.

[5M]L2

OR

3. a) Compare quantitative and qualitative data representations.

[5M]L4

b) Define data preprocessing. Explain common methods of data preprocessing.

[5M]L2

4. a) Discuss different forms of Taxonomies in Visualization.

[5M]L2

b) Interpret Experimental Semiotics based on Perception.

[5M]L3

OR

5. a) Illustrate eight visual variables used for data visualization.

[5M]L2

b) Illustrate three stage model of perceptual processing.

[5M]L2

6. Explain with respect to geospatial data: [10M]L3
a) Choropleth maps
b) Cartograms

OR

7. a) Explain Scatterplots and Scatterplot Matrices for Point-Based multivariate visualization. [5M]L2
b) Compare line-based techniques in visualization of multivariate data. [5M]L2

8. a) Compare the different interaction operators in data visualization [5M]L4
b) Explain unified framework in Interaction Concepts. [5M]L2

OR

9. a) Analyze tasks using vector space model. [5M]L4
b) Write a short note on arc diagrams and word tree visualization. [5M]L2

10. a) Discuss problems in designing effective Visualizations. [5M]L2
b) Explain the Importance of Aesthetics in Visualization. [5M]L2

OR

11. a) Explain briefly Issues of Applications in System-Design. [5M]L2
b) Discuss various issues of Hardware and Applications in Visualization [5M]L2

---ooOoo---



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500029

B.Tech IV Year I Semester Supplementary Examinations, June-2025

DATA VISUALIZATION TECHNIQUES

(COMMON TO CSE, CSE(AI&ML), CSE(DS))

Time:3 hours

Max. Marks:70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART-A

(20Marks)

1. a) Define data visualization. [2M]L1
- b) List the types of data used in visualization. [2M]L1
- c) Name the eight visual variables. [2M]L1
- d) Define semiology of graphical symbols. [2M]L1
- e) What is spatial data? [2M]L1
- f) What is multi variate data? [2M]L1
- g) What is the vector space model in document visualization? [2M]L1
- h) List interaction operators used in visualizations. [2M]L1
- i) What are the issues of cognition in visualization design? [2M]L1
- j) List common problems in designing visualizations. [2M]L1

PART-B

(5X10=50Marks)

2. Analyze the steps involved in the visualization process. [10M]L4

OR

3. Explain the importance of Data preprocessing before visualization. [10M]L2
4. Explain the importance of visual variables in the design of graphics. [10M]L4

OR

5. Justify the need for experimental semiotics in visualization design. [10M]L4
6. Evaluate the challenges in displaying hierarchical structures using tree graphics. [10M]L3

OR

7. Compare visualization methods for point data and area data in case of Geo spatial data. [10M]L4

8. Explain the role of interaction operands and spaces in visualization. [10M]L2

OR

9. Assess the effectiveness of different interaction techniques in enhancing user experience. [10M]L4

10. Analyze the influence of perception and cognition on visualization effectiveness. [10M]L4

OR

11. Evaluate the criteria for measuring the quality of a designed visualization. [10M]L3

---ooOoo---

KR21

Code No: 217AD



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Regular Examinations, January-2025

DATA VISUALIZATION TECHNIQUES

(COMMON TO CSE (AI&ML), CSE (DS))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(20 Marks)

1. a) What are the tools used in computer graphics for visualizing data and images? [2M] L1
- b) Give examples for data visualization. [2M] L1
- c) List out visual variables used in data visualization? [2M] L1
- d) What is the use of colour as a visual variable? [2M] L1
- e) Define Geospatial Data visualization. [2M] L1
- f) Mention the uses of choropleth maps? [2M] L1
- g) List out different interaction operators. [2M] L1
- h) What is meant by lexical level of text representation. [2M] L1
- i) What is the importance of information density in designing a visualization? [2M] L1
- j) What are the key issues associated with research directions in visualizations? [2M] L1

PART- B

(5 X 10 =50 Marks)

2. a) Explain Computer Graphics pipeline. [5M] L2
- b) Describe how data visualization is related to computer science and information technology [5M] L2

OR

3. a) Write a short notes on Pseudo-code conventions and scatter plots. [5M] L2
- b) Define Data Preprocessing. Explain common methods of Data Preprocessing. [5M] L2
4. a) Explain the semiology of graphical symbols and their significance in data visualization. [5M] L2
- b) Discuss the stages of visualization and illustrate them with a neat diagram. [5M] L2

OR

5. a) Summarize the different forms of taxonomies used in visualization. [5M] L2
 b) Explain briefly about historical perspective of visualization. [5M] L2

6. a) Discuss the different techniques used for the visualization of line data in geospatial contexts, highlighting their significance and application. [5M] L2
 b) Explain two dimensional spatial data with respect to data visualization. [5M] L2

OR

7. a) Explain Scatterplots and Scatterplot Matrices in the context of Point-Based Techniques in visualization of Multivariate Data. [5M] L2
 b) What are the different issues in geospatial data visualization? [5M] L2

8. a) What are the different levels of text representation? Explain. [5M] L2
 b) Explain document collection visualization. [5M] L2

OR

9. a) Explain Interaction Operands and Spaces. [5M] L2
 b) List out the different interaction techniques. Explain [5M] L2

10. a) Write a short notes on
 I) Intuitive mappings from data to visualization
 II) Selecting and modifying views [5M] L2
 b) List out issues of Hardware and Applications in visualization. Explain [5M] L2

OR

11. a) Differentiate Spatial vs Non Spatial Data vs Dynamic Data [5M] L2
 b) Compare issues of Cognition, Perception and Reasoning in visualization [5M] L4

---ooOoo---

Code No: 217AD

KR21**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Regular Examinations, January-2025**DATA VISUALIZATION TECHNIQUES
(COMPUTER SCIENCE AND ENGINEERING)****Time: 3 hours****Max. Marks: 70****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A**(20 Marks)**

1. a) Define Data Visualization and explain its importance. [2M] L2
- b) What is meant by data preprocessing? [2M] L1
- c) Define taxonomy in the context of visualization. [2M] L1
- d) What is meant by perpetual processing. [2M] L1
- e) Explain the importance of contour map. [2M] L2
- f) Write a short note on point based techniques with respect to multivariate data. [2M] L1
- g) What is the use of navigation operators? Illustrate [2M] L2
- h) Write a brief note on Vector space model. [2M] L1
- i) Differentiate structured data from non-structured data [2M] L2
- j) What are the issues of cognition? [2M] L1

PART- B**(5 X 10 =50 Marks)**

2. Illustrate the Visualization process in detail. [10M] L3

OR

3. a) Distinguish Data Visualization from computer graphics. [5M] L2
- b) Summarize structures within and between records related to datasets. [5M] L2
4. a) Describe Gibson's affordance theory with respect to perception. [5M] L2
- b) Write a short notes on (i) Symbols and visualization (ii) Features of Graphics [5M] L2

OR

5. a) Compare eight visual variables used for data visualization in detail. [5M] L4
- b) Illustrate the three stage model of perceptual processing. [5M] L3

6. a) Discuss the various techniques for visualization of point data related to geospatial data. [5M] L2
b) Compare line based techniques in visualization of multivariate data. [5M] L4

OR

7. a) Differentiate space filling methods from non-space filling methods in displaying hierarchical structures. [5M] L3
b) Defining spatial data visualization, explain about one dimensional spatial data with respect to data visualization techniques. [5M] L2

8. a) Demonstrate Zipf's law and describe its importance. [5M] L2
b) Illustrate different types of visualization of a single document. [5M] L3

OR

9. a) Compare the different interaction operators in data visualization. [5M] L4
b) Explain the parameters required for Unified framework. [5M] L2

10. a) Describe in detail the issues of Cognition. [5M] L2
b) Elaborate the problems in designing effective visualizations. [5M] L2

OR

11. Discuss the steps in designing visualizations. [10M] L2

---ooOoo---

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Regular Examinations, December -2023

DATA VISUALIZATION TECHNIQUES
(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 75

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1. a) What is visualization? [2M] L1
- b) Discuss the relationship between visualization and human-computer interaction. [3M] L3
- c) List out Visualization Stages. [2M] L2
- d) Explain the visualization process in detail. [3M] L2
- e) What are Visualization Techniques for Spatial Data for One-Dimensional Data? [2M] L1
- f) How is spatial data visualized? [3M] L1
- g) What is a Unified Framework? [2M] L1
- h) Explain Object-Space for 3D Surfaces. [3M] L2
- i) What are issues of Data? [2M] L1
- j) What are the steps in designing the data virtualization? [3M] L1

PART- B

(5 X10=50Marks)

2. a) Explain i) The Visualization Process ii) Pseudo code Conventions iii) The Scatter plot [6M] L2
- b) Explain types of data in Data Foundation. [4M] L2

OR

3. a) What is data set? How do data sets contribute to data analysis? [5M] L4
- b) Discuss about Structure within and between Records. [5M] L2

4. a) Explain the Eight Visual Variables used for graphical objects in visualization. [5M] L2
- b) Explain the historical perspective and taxonomies. [5M] L2

OR

5. a) Illustrate experimental semiotics based on perception Gibson's affordance theory. [5M] L4
- b) Explain a model of perceptual processing. [5M] L2

6. a) Discuss about Two-Dimensional spatial Data. [5M] L2
b) Elaborate on visualization techniques of point data for geospatial data. [5M] L2

OR

7. a) Discuss the Visualization Techniques for Multivariate Data using Point-Based. [5M] L4
b) Explain the line integral convolution. [5M] L2

8. a) Explain about the levels of text representations in Text and Document Visualization. [5M] L2
b) Discuss about the Vector Space Model. [5M] L2

OR

9. a) Identify the Interaction Operators. Explain. [4M] L1
b) Explain about Interaction Operands and Spaces. [6M] L2

10. a) Discuss the Problems in designing effective Visualizations. [4M] L5
b) Discuss the Issues of i) Cognition ii) Perception, and iii) Reasoning. [6M] L4

OR

11. a) Explain the issues of System Design. [5M] L4
b) Explain the issues of Evaluation during designing effective visualizations. [5M] L2

---ooOoo---



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Supplementary Examinations, May -2024

DATA VISUALIZATION TECHNIQUES

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 75

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1. a) What is the relationship between visualization and other fields such as computer science and medical science. [2M] L2
- b) What is the importance of scatter plots in visualizing data relationships? [3M] L1
- c) List any four Visual Variables. [2M] L1
- d) What is Gibson's Affordance theory? [3M] L1
- e) Define 1-D Spatial data. [2M] L1
- f) Explain the two approaches of graph visualization. [3M] L2
- g) What are Levels of Text Representations? [2M] L1
- h) Define Attribute Space and Data Structure Space. [3M] L1
- i) What are the challenges in designing effective Visualization? [2M] L2
- j) List any four Latest Data Visualization tools. [3M] L1

PART- B

(5 X 10 =50 Marks)

2. Given a dataset containing various types of data (numerical, categorical, textual), design and implement a data preprocessing pipeline. Describe the techniques used for each type of data and justify their selection. [10M] L3

OR

3. a) Explain nominal data and ordinal data and give an example of each one. [5M] L2
- b) Is Data visualization required? Justify your answer. [5M] L4
4. a) Analyze the role of historical perspectives in shaping modern data Visualization techniques. [5M] L4
- b) What are the semiotic principles for representing different types of data? Explain. [5M] L2

OR

5. a) Explain stages in data visualization. [5M] L2
b) Explain the rationale behind the design choices and demonstrate how each symbol effectively conveys information. [5M] L2

6. a) Describe the design considerations and interaction features that enhance user engagement and data exploration. [5M] L3
b) Compare and contrast different visualization techniques for representing multivariate data. [5M] L4

OR

7. a) Explain the technique for visualizing spatial data in three dimensions. [5M] L2
b) Discuss potential issues related to data integration, visual coherence, and user comprehension. [5M] L2

8. Evaluate the impact of different interaction spaces (e.g., screen space, data space) on user engagement and comprehension in data visualization. Discuss how the choice of interaction space influences the effectiveness of visualization techniques. [10M] L4

OR

9. a) Explain any 4 interaction operators. [5M] L2
b) Explain Interaction Control in detail. [5M] L2
10. Evaluate the effectiveness of existing evaluation methods for assessing the usability and utility of data visualization systems. Discuss their strengths and limitations, and propose improvements or alternative approaches to address identified shortcomings. [10M] L4

OR

11. a) Outline the research objectives, methodology, and expected outcomes of the study "Research Directions in Virtualizations". [5M] L2
b) Discuss Issues of System Design Evaluation. [5M] L2

---ooOoo---

Code No: 227AD

RKR21



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE; Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Regular Examinations, December-2025/January-2026

DATA VISUALIZATION TECHNIQUES

(COMMON TO CSE (AI&ML), CSE (DS))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(20 Marks)

1. a) What is Data Visualization.

[2M]L1

b) Differentiate Data Visualization and Computer Graphics.

[2M]L2

c) Write short note on Bertins Semiology of Graphics.

[2M]L1

d) Why is contrast important in visualization?

[2M]L2

e) Write short notes on Probing Two-Dimensional Data

[2M]L1

f) Explain Slicing techniques in Visualizing Volume Data.

[2M]L2

g) What are the advantages of visualizing a single document using techniques like word clouds or tag clouds?

[2M]L2

h) List any four Interaction Techniques.

[2M]L1

i) What is the importance of information density in designing a visualization?

[2M]L1

j) How do limitations in human perception influence visualization design?

[2M]L2

PART- B

(5 X 10 =50 Marks)

2. a) How data visualization is related to computer science and information technology?

[5M]L2

b) Explain the significance of handling missing data and describe common techniques.

[5M]L2

OR

3. a) Compare quantitative and qualitative data representations.

[5M]L4

b) Define data preprocessing. Explain common methods of data preprocessing.

[5M]L2

4. a) Discuss different forms of Taxonomies in Visualization.

[5M]L2

b) Interpret Experimental Semiotics based on Perception.

[5M]L3

OR

5. a) Illustrate eight visual variables used for data visualization.

[5M]L2

b) Illustrate three stage model of perceptual processing.

[5M]L2

6. Explain with respect to geospatial data:

[10M]L3

- a) Choropleth maps
- b) Cartograms

OR

7. a) Explain Scatterplots and Scatterplot Matrices for Point-Based multivariate visualization.

[5M]L2

b) Compare line-based techniques in visualization of multivariate data.

[5M]L2

8. a) Compare the different interaction operators in data visualization

[5M]L4

b) Explain unified framework in Interaction Concepts.

[5M]L2

OR

9. a) Analyze tasks using vector space model.

[5M]L4

b) Write a short note on arc diagrams and word tree visualization.

[5M]L2

10. a) Discuss problems in designing effective Visualizations.

[5M]L2

b) Explain the Importance of Aesthetics in Visualization.

[5M]L2

OR

11. a) Explain briefly Issues of Applications in System-Design.

[5M]L2

b) Discuss various issues of Hardware and Applications in Visualization

[5M]L2

---ooOoo---

**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Regular Examinations, December-2025/January-2026

DATA VISUALIZATION TECHNIQUES

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(20 Marks)

1. a) Define data preprocessing. Explain common methods of data preprocessing. [2M]L1
- b) What is the difference between quantitative and categorical data in the context of data foundations. [2M]L2
- c) Name any four visual variables used for encoding data. [2M]L2
- d) Explain the importance of labeling and annotation in graphs. [2M]L1
- e) Differentiate between bar charts and line charts. [2M]L2
- f) Summarize point phenomena, line phenomena in Spatial Data. [2M]L2
- g) State Zipf's law in visualization. [2M]L2
- h) Mention any two levels of text representation and explain their significance. [2M]L1
- i) Describe two system design issues that affect the scalability of visualization tools. [2M]L2
- j) List the key issues in Research Directions in visualization. [2M]L2

PART- B

(5 X 10 =50 Marks)

2. a) Describe the scatterplot in detail. Explain its components, uses, limitations. [5M]L3
- b) Give important steps in computer graphics pipeline. [5M]L2

OR

3. a) Define data visualization. Discuss the importance of data visualization. [5M]L2
- b) Summarize structures within and between records related to data sets. [5M]L2

4. a) Explain in detail Historical Perspective of visualization. [5M]L2
- b) Describe Gibson's Affordance theory with respect to perception. [5M]L2

OR

5. a) Illustrate Visualization stages with a neat diagram. [5M]L4
- b) Explain how designers choose visual encoding for different datasets. [5M]L3

6. a) Explain different 1D data visualization techniques with examples. [5M]L2

b) Write a short note on parallel coordinates and Andrews Curves (line based techniques for multivariate data). [5M]L4

OR

7. a) What are the various Visual Variables for Spatial Data? Explain with a neat diagram. [5M]L2

b) Illustrate basic visualization techniques for three-dimensional data. [5M]L4

8. a) Explain different levels of text representation. [5M]L2

b) Compare the different interaction operators in data visualization. [5M]L4

OR

9. a) Illustrate different types of visualization of a single text document. [5M]L3

b) Discuss Self-Organizing Maps and Themescapes in Document Collection Visualization. [5M]L2

10. a) Compare Issues of Cognition, Perception, and Reasoning in visualization. [5M]L4

b) Discuss various issues of Hardware and Applications in Visualization. [5M]L2

OR

11. a) Write short notes on Intuitive Mappings from Data to Visualization. [5M]L2

b) Categorize Issues of Data with respect to research directions in Visualization. [5M]L4

---ooOoo---